UAV Launch System

Actuator Component Selection

Full-Scale System and Scaled Test Model

Component selection and design rationale

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1 Purpose and Design Approach

This document records the selected actuator components for the UAV launch system, including the full-scale assembly and the scaled test assembly. It summarizes the assumed system dimensions and masses, the required actuator performance obtained from the `actuatorValues.py` file, the selected motor and Acme lead-screw components, and the reasoning behind each selection.

The calculations and derivations are not repeated here. The equations, free-body diagram, assumptions, and original sizing calculations are provided in the companion document "UAV Launch Actuator Values". The broader comparison of actuator technologies and their advantages and disadvantages is provided in "Actuator Pros and Cons".

The original design assumed the use of two linear actuators to move the two desired rails individually. This was intended to reduce the required force per actuator. However, to achieve the required motor RPM, the available servo motors meet the requirements for a single actuator moving both rails. This also reduces the problem of potentially actuating at different rates, which could cause torque on the launcher and plane.

One possible option at this scale would be to use the selected motors at the base, accompanied by a reduction gearbox to achieve the required torque. However, as the test model is a proof of concept, and increasing the scale would require longer rails, this approach would quickly become more expensive and difficult to implement than using linear actuators near the end of the rails. For this reason, it was not considered for this project.

1.1 Selected Actuator Architecture

Both configurations use a motor-driven Acme lead screw to convert rotary motion into linear motion. The lead screw was chosen instead of ball screws because the required motor torque was not high enough to justify the need for a higher efficiency coefficient to meet the required force.

The lead-screw nut is attached to the moving structure, while the screw is driven by an integrated servo motor. Motor control and powering are not considered in this documentation.

The selected StepperOnline motors are integrated servo units with an internal drive and encoder. A separate external motor driver is therefore not intended for this arrangement.

1.2 Selection Criteria

The principal selection criteria are:

- Sufficient motor torque at the required screw speed.
- Sufficient linear thrust and travel speed for the required motion.
- Compatibility between the screw and nut thread.
- A practical mechanical connection between the motor shaft and screw.
- Practicality of fabrication, assembly, testing, and maintenance.

2 Full-Scale Actuator System

2.1 Assumed System Parameters

The following values are the full-scale system assumptions supplied for the actuator sizing. The value of the beam mass is shown both as entered and as a total mass.

Parameter	Description	Value
L_1	Beam length	2.0 m
L_2	Attached-object length	1.5 m
m_1	Beam mass, entered as	3.34 kg
m_2	Attached-object mass	11.0 kg
r_f	Actuator attachment distance from pivot	1.75 m
$\Delta\theta$	Required angular displacement	10°
T	Desired motion time	0.5 s
g	Gravitational acceleration	9.81 m/s^2
L_s	Lead used in the original sizing calculation	0.012 m/rev
η	Assumed screw efficiency in the original sizing calculation	0.60

Table 1: Full-scale system assumptions.

The values for attached-object length and mass are taken from the MakeFlyEasy Fighter Hand Throw specifications. The values for beam length, required angular displacement, and desired motion time were taken from the problem statement. Gravity and assumed screw efficiency were treated as constants.

The actuator attachment distance and lead used were initially arbitrary, then varied to find a position and lead that provided suitable values for motor selection, e.g., a balance between required RPM and torque. The values shown were ultimately those that best fit the motor capabilities.

2.2 Required Actuator Performance

The following results are transcribed from the supplied full-scale sizing calculation. They are retained here as the original calculated requirements.

Calculated quantity	Original result
Angular acceleration	2.793 rad/s^2
Maximum required actuator force	85.241 N
Recommended actuator force (safety factor of 2.0)	170.483 N
Linear travel reported by the calculation	0.2618 m
Average linear speed reported by the calculation	0.6109 m/s
Motor torque reported by the calculation	0.5427 N m
Screw speed reported by the calculation	3054.3 RPM

Table 2: Original full-scale actuator sizing results.

2.3 Selected Full-Scale Components

Component	Selected part / relevant specification	Supplier link
Motor	StepperOnline iSV57T-180S integrated servo; rated power 180 W, rated torque 0.6 N m, rated speed 3000 RPM, input range 20–50 VDC, nominal 8 mm shaft.	StepperOnline iSV57T-180S
Lead screw	McMaster-Carr 98980A973; 3/8 in–8 Stub Acme, four-start, 0.5 in lead, 3 ft length, 304 stainless steel.	McMaster-Carr 98980A973
Lead nut	McMaster-Carr 1349K101; matching 3/8 in–8 Acme, four-start, 0.5 in lead, PET nut; catalog dynamic thrust rating 350 lbf.	McMaster-Carr 1349K101
Shaft coupler	Abra Electronics / ServoCity listing supplied for a 1/4 in to 8 mm set-screw coupler. The screw is intended to be machined to a 1/4 in smooth journal for the coupler.	Abra coupler listing

Table 3: Selected full-scale components and supplier links.

2.4 Full-Scale Motor Selection Rationale

The selected motor, lead screw, and nut combination was decided upon based on the given values and desired functionality. The original calculation requires 0.3172 N m at 2618 RPM, while the motor is listed at 0.6 N m rated torque and 3000 RPM rated speed, with a 1.1 N m maximum torque and 4000 RPM maximum speed. These values are above what is needed according to the calculation, which already includes a 2X safety factor. The motor torque–speed curve is shown below.

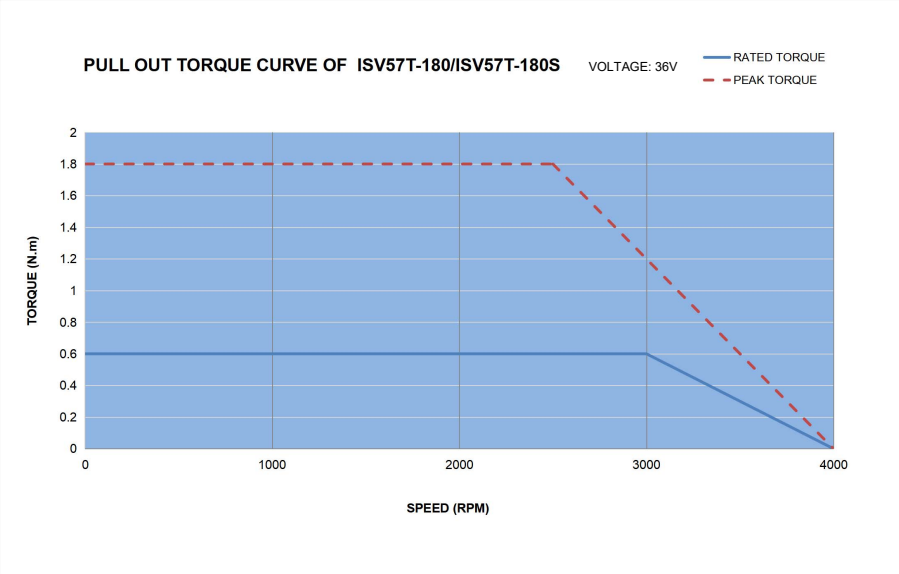


Figure 1: Manufacturer torque–speed curve for the selected StepperOnline iSV57T-180S integrated servo motor.

2.5 Full-Scale Lead Screw and Nut Rationale

The selected screw and nut are a matched 3/8 in–8 Acme pair with a four-start thread and a nominal lead of 0.5 in per revolution. This corresponds to 12.7 mm/rev. The PET nut is catalogued with a 350 lbf dynamic thrust rating. These components provide the rotary-to-linear transmission for the full-scale actuator.

2.6 Full-Scale Coupling and Machining

The motor has a nominal 8 mm shaft. The supplied coupler link is identified as a 1/4 in to 8 mm coupler. The intended manufacturing approach is to machine the lead-screw end to a concentric 1/4 in smooth journal and secure it using the coupler set screw.

3 Scaled Test Actuator System

3.1 Assumed Test-System Parameters

The scaled test model uses reduced beam and object masses and a shorter actuator attachment radius. The angular displacement and desired motion time remain the same as those specified for the full-scale system.

Parameter	Description	Value
L_1	Beam length	1.0 m
L_2	Attached-object length	0.5 m
m_1	Beam mass	1.67 kg
m_2	Attached-object mass	5.0 kg
r_f	Actuator attachment distance from pivot	0.5 m
$\Delta\theta$	Required angular displacement	10°
T	Desired motion time	0.5 s
g	Gravitational acceleration	9.81 m/s^2
L_s	Lead used in the original sizing calculation	0.006 m/rev
η	Assumed screw efficiency in the original sizing calculation	0.60

Table 4: Scaled test-system assumptions.

The values for attached-object length and mass, beam length, required angular displacement, and desired motion time were taken from the problem statement. Gravity and assumed screw efficiency were treated as constants.

The actuator attachment distance and lead used were initially arbitrary, then varied to find a position and lead that provided suitable values for motor selection, e.g., a balance between required RPM and torque. The values shown were ultimately those that best fit the motor capabilities while also fitting within the testing budget.

3.2 Required Actuator Performance

The following results are transcribed from the supplied scaled-model sizing calculation.

Calculated quantity	Original result
Angular acceleration	2.793 rad/s^2
Maximum required actuator force	46.344 N
Recommended actuator force (safety factor of 2.0)	92.688 N
Linear travel reported by the calculation	0.0873 m
Average linear speed reported by the calculation	0.1745 m/s
Motor torque reported by the calculation	0.1475 N m
Screw speed reported by the calculation	1745.3 RPM

Table 5: Original scaled-model actuator sizing results.

3.3 Selected Test-System Components

Component	Selected part / relevant specification	Supplier link
Motor	StepperOnline iSV57T-090 integrated servo; rated power 90 W, rated torque 0.3 N m, rated speed 3000 RPM, input range 20–50 VDC, nominal 8 mm shaft.	StepperOnline iSV57T-090
Lead screw	McMaster-Carr 99030A986; 3/8 in–8 Stub Acme, two-start, 0.25 in lead, 18 in length, 1018 low-carbon steel.	McMaster-Carr 99030A986
Lead nut	McMaster-Carr 1349K107; matching 3/8 in–8 Acme, two-start, 0.25 in lead, PET nut; catalog dynamic thrust rating 350 lbf.	McMaster-Carr 1349K107
Shaft coupler	Abra Electronics / ServoCity listing supplied for a 5/16 in to 3/8 in set-screw coupler. The screw is intended to be welded or temporarily fixed for the test system.	Abra coupler listing

Table 6: Selected scaled test-system components and supplier links.

3.4 Test Motor Selection Rationale

The selected motor, lead screw, and nut combination was decided upon based on the given values and desired functionality. The motor's listed rated torque and speed provide a nominal comparison against the original calculated demand of 0.1475 N m at 1745.3 RPM. The motor is listed at 0.3 N m rated torque and 3000 RPM rated speed, with a 0.8 N m maximum torque and 4000 RPM maximum speed. These values are above what is needed according to the calculation, which already includes a 2X safety factor. The motor torque–speed curve is shown below.

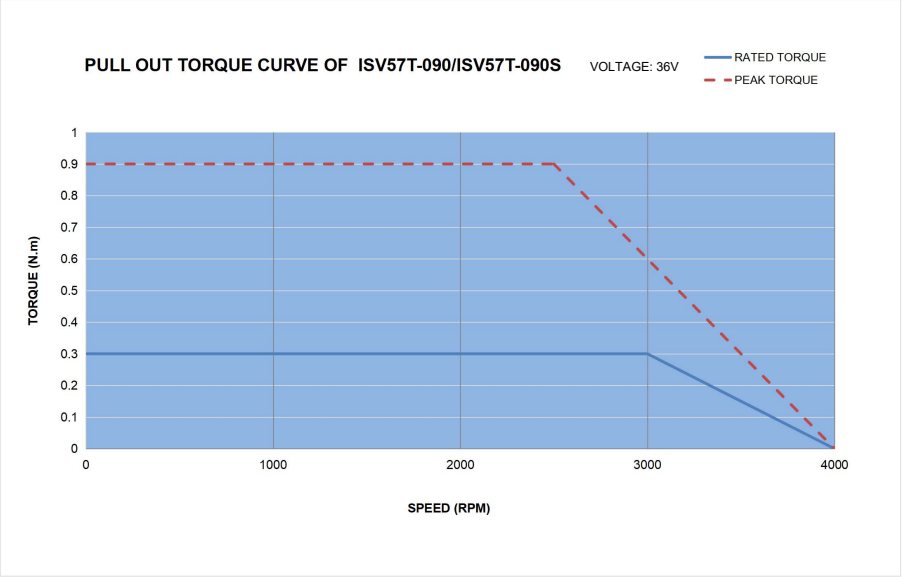


Figure 2: Manufacturer torque–speed curve for the selected StepperOnline iSV57T-090 integrated servo motor.

3.5 Test Lead Screw, Nut, and Coupling Rationale

The selected screw and nut form a matching 3/8 in–8 Acme pair with a two-start thread and a nominal lead of 0.25 in per revolution. This corresponds to 6.35 mm/rev. The PET nut is catalogued with a 350 lbf dynamic thrust rating.

The supplied coupler is listed as 5/16 to 3/8. The intended manufacturing approach is to temporarily fix or weld the lead screw to the coupler, as a means of minimizing manufacturing costs associated with machining.

4 Selection Summary

Item	Full-scale system	Scaled test system
Motor	iSV57T-180S, 180 W, 0.6 N m rated torque	iSV57T-090, 90 W, 0.3 N m rated torque
Screw	98980A973, 3/8 in–8, four-start Acme, 0.5 in lead	99030A986, 3/8 in–8, two-start Acme, 0.25 in lead
Nut	1349K101, matching PET nut	1349K107, matching PET nut
Original design thrust	99.652 N	92.688 N
Original calculated torque	0.3172 N m	0.1475 N m
Original calculated speed	2618.0 RPM	1745.3 RPM
Control / power	Arduino Uno STEP/DIR; external nominal 36 V battery	Arduino Uno STEP/DIR; external nominal 36 V battery

Table 7: Summary of the selected components and original calculated requirements.

The selected components meet the required specifications for the intended tasks, both for the full-scale system and the scaled testing model.

References

[1] STEPPERONLINE, "NEMA 23 Integrated Easy Servo Motor 180W, 3000 RPM, 0.6 N·m, iSV57T-180S." [Online]. Available: https://www.stepperonline.ca/nema-23-integrated-easy-servo-motor-180w-3000rpm-0-6nm-84-98oz-in-20-50vdc-servo-motor-short-shaft-isv57t-180s.html. [Accessed: Nov. 12, 2026].

[2] STEPPERONLINE, "NEMA 23 Integrated Easy Servo Motor 90W, 3000 RPM, 0.3 N·m, iSV57T-090." [Online]. Available: https://www.stepperonline.ca/nema-23-integrated-easy-servo-motor-90w-3000rpm-0-3nm-42-49oz-in-20-50vdc-brushless-dc-servo-motor-isv57t-090.html. [Accessed: Nov. 12, 2026].

[3] McMaster-Carr, "Acme Lead Screw, Part No. 98980A973." [Online]. Available: https://www.mcmaster.com/98980A973/. [Accessed: Nov. 12, 2026].

[4] McMaster-Carr, "Acme Lead Screw Nut, Part No. 1349K101." [Online]. Available: https://www.mcmaster.com/1349K101/. [Accessed: Nov. 12, 2026].

[5] Abra Electronics, "1/4 in. to 8 mm Set Screw Shaft Coupler, Part No. 625182." [Online]. Available: https://abra-electronics.com/robotics-actobotics-servocity/motion-en/couplers/set-screw-shaft-couplers/625182-1-4-to-8mm-set-screw-shaft-coupler-actobotics-clone.html. [Accessed: Nov. 12, 2026].

[6] McMaster-Carr, "Acme Lead Screw, Part No. 99030A986." [Online]. Available: https://www.mcmaster.com/99030A986/. [Accessed: Nov. 12, 2026].

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[8] Abra Electronics, "5/16 in. to 5/16 in. Set Screw Shaft Coupler, Part No. 625126." [Online]. Available: https://abra-electronics.com/robotics-actobotics-servocity/motion-en/couplers/set-screw-shaft-couplers/625126-5-16-to-5-16-set-screw-shaft-coupler-actobotics-clone.html. [Accessed: Nov. 12, 2026].

[9] "UAV Launch Actuator Values and Calculations," unpublished project document, 2026.

[10] "Actuator Pros and Cons," unpublished project document, 2026.